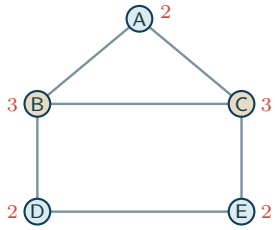
GRAPH THEORY LADDER 1 OF 9

Graphs: vertices, edges and degrees

A graph records which things are linked, and counting the links at each vertex leads to the handshake rule.

1 / THE RULE AND THE PICTURE

Adding up the degrees counts every edge twice



the house graph, degrees in red

A graph is a set of vertices joined by edges, and the degree of a vertex is the number of edges that meet it. Every edge has two ends, so adding the degrees counts each edge twice:

$$\boxed{\sum_v \deg v = 2E}$$

For the house graph the degrees $2, 3, 3, 2, 2$ add to 12, twice its 6 edges. This graph is the running example of the ladder: it has 5 vertices, 6 edges and one triangle.

2 / SOME STANDARD GRAPHS

A few families appear everywhere

graph	vertices	edges	description
path P_n	n	$n-1$	a line of n vertices
cycle C_n	n	n	a ring of n vertices, every degree 2
complete graph K_n	n	$\frac{n(n-1)}{2}$	every pair joined, K_5 has 10 edges
complete bipartite $K_{m,n}$	$m+n$	mn	every vertex of one group joined to every vertex of the other, $K_{3,3}$ has 9
star $K_{1,n}$	$n+1$	n	one centre joined to n leaves

3 / PARTY PUZZLES

The sum of the degrees is always even

Five people cannot each be friends with exactly three others, since the degrees would add to 15, which is odd and so cannot equal $2E$.
For the same reason the number of vertices of odd degree is always even, and in any gathering an even number of people have shaken an odd number of hands.
The degrees do not settle everything, though: a ring of six vertices and two separate triangles have six vertices of degree 2 each, yet the first is connected and the second is not.

4 / WHAT GRAPHS MODEL

Choose the vertices, then decide what an edge means

The vertices can be towns with roads as edges, people with friendships, web pages with links, or atoms with bonds: the carbon skeleton of butane is a path P_4 and the benzene ring is a cycle C_6 with every degree 2.
The picture throws away the lengths and the angles and keeps only the pattern of links, which is why one graph can stand for many different physical situations.

5 / TRY THESE

Count with the handshake rule

(a) How many edges has K_6? Each of 6 vertices has degree 5, so $6 \times 5/2 = 15$.
(b) A graph has degrees $3, 3, 2, 2, 2$: how many edges? $12/2 = 6$.
(c) Can a graph have degrees $3, 3, 3$? Their sum is 9, which is odd, so no.

ASIDE / SIMPLE GRAPHS

A simple graph has no loops and at most one edge between two vertices; the bridges of Königsberg later need repeated edges.

LINKS TO OTHER SHEETS

The next sheet follows paths through the house graph; the carbon chain sheet in the physics collection uses the same graphs for molecules.

Convention. V vertices, E edges, $\deg v$ the degree of v; P_n, C_n, K_n, $K_{m,n}$ are the path, cycle, complete and complete bipartite graphs.